Solutions to Axler's Linear Algebra

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Contents

1 Vector Spaces

1A $\mathbb{R}^n$ and $\mathbb{C}^n$

Exercise 1A1

Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$ and $\beta = c + di$ for some $a, b, c, d \in \mathbb{R}$. Then

$$\begin{aligned}\alpha + \beta &= (a + bi) + (c + di) \\ &= (a + c) + (b + d)i \\ &= (c + a) + (d + b)i \\ &= (c + di) + (a + bi) = \beta + \alpha. \end{aligned}$$

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Exercise 1A2

Show that $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}(\alpha + \beta) + \gamma &= ((a + bi) + (c + di)) + (e + fi) \\ &= ((a + c) + (b + d)i) + (e + fi) \\ &= (a + c + e) + (b + d + f)i \\ &= (a + (c + e)) + (b + (d + f))i \\ &= (a + bi) + ((c + di) + (e + fi)) = \alpha + (\beta + \gamma). \end{aligned}$$

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Exercise 1A3

Show that $(\alpha\beta)\gamma = \alpha(\beta\gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}(\alpha\beta)\gamma &= ((a + bi)(c + di))(e + fi) \\ &= ((ac - bd) + (ad + bc)i)(e + fi) \\ &= ((ac - bd)e - (ad + bc)f) + ((ac - bd)f + (ad + bc)e)i \\ &= (a(ce) - b(de) - a(df) - b(cf)) + (a(cf) - b(df) + a(de) + b(ce))i \\ &= \alpha(\beta\gamma). \end{aligned}$$

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Exercise 1A4

Show that $\gamma(\alpha + \beta) = \gamma\alpha + \gamma\beta$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}\gamma(\alpha + \beta) &= (e + fi)((a + bi) + (c + di)) \\ &= (e + fi)((a + c) + (b + d)i) \\ &= (e(a + c) - f(b + d)) + (e(b + d) + f(a + c))i \\ &= (ea - fb + ec - fd) + (eb + fa + ed + fc)i \\ &= (ea - fb) + (eb + fa)i + (ec - fd) + (ed + fc)i \\ &= (e + fi)(a + bi) + (e + fi)(c + di) = \gamma\alpha + \gamma\beta. \end{aligned}$$

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Exercise 1A5

Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Solution. Let $\alpha = a + bi$ for some $a, b \in \mathbb{R}$. We want to find $\beta = c + di$ such that $\alpha + \beta = 0$. This gives

$$(a + bi) + (c + di) = (a + c) + (b + d)i = 0 + 0i.$$

Solving for c and d, we get $c = -a$ and $d = -b$. Hence, $\beta = c + di = -a - bi = -\alpha$.

To see that β is unique, suppose there is another $\gamma \in \mathbb{C}$ such that $\alpha + \gamma = 0$. Then

$$\beta = \beta + 0 = \beta + (\alpha + \gamma) = (\beta + \alpha) + \gamma = 0 + \gamma = \gamma.$$

Hence, β is unique. ■

Exercise 1A6

Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Solution. Let $\alpha = a + bi$ for some $a, b \in \mathbb{R}$ with $\alpha \neq 0$. We want to find $\beta = c + di$ such that $\alpha\beta = 1$. This gives

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i = 1 + 0i.$$

Solving for c and d, we get the system of equations

$$\begin{cases} ac - bd = 1 \\ ad + bc = 0 \end{cases} \implies \begin{cases} a^2c - abd = a \\ abd + b^2c = 0 \end{cases}$$

Adding the two equations, we get

$$(a^2 + b^2)c = a \implies c = \frac{a}{a^2 + b^2}.$$

Substituting this into the second equation, we get

$$abd + b^2\frac{a}{a^2 + b^2} = 0 \implies d = \frac{-b}{a^2 + b^2}.$$

Hence,

$$\beta = c + di = \frac{a}{a^2 + b^2} + \frac{-b}{a^2 + b^2}i.$$

If $\gamma \in \mathbb{C}$ is another complex number such that $\alpha\gamma = 1$, then

$$\gamma = \gamma 1 = \gamma(\alpha\beta) = (\gamma\alpha)\beta = 1\beta = \beta.$$

Hence, β is unique. ■

Exercise 1A7

Show that

$$\frac{-1 + \sqrt{3}i}{2}$$

is a cube root of 1.

Solution. Note that the complex number given can be written as

$$\frac{1}{2}(-1 + \sqrt{3}i).$$

Cubing $-1+\sqrt{3}i$, we get

$$\begin{aligned}(-1+\sqrt{3}i)^3 &= (-1)^3 + 3(-1)^2(\sqrt{3}i) + 3(-1)(\sqrt{3}i)^2 + (\sqrt{3}i)^3 \\ &= -1 + 3\sqrt{3}i + 3(-1)(-3) + (\sqrt{3}i)^3 \\ &= -1 + 3\sqrt{3}i + 9 + 3\sqrt{3}i(-i) \\ &= -1 + 3\sqrt{3}i + 9 - 3\sqrt{3}i \\ &= 8.\end{aligned}$$

Therefore,

$$\left(\frac{-1+\sqrt{3}i}{2}\right)^3 = \frac{(-1+\sqrt{3}i)^3}{8} = \frac{8}{8} = 1. \qquad \blacksquare$$

Exercise 1A8

Find two distinct square roots of i.

Solution. Let $z = a + bi$ for some $a, b \in \mathbb{R}$. We want to find z such that $z^2 = i$. This gives

$$(a+bi)^2 = a^2 - b^2 + 2abi = 0 + 1i.$$

Solving for a and b, we get the system of equations

$$\begin{cases} a^2 - b^2 = 0 \\ 2ab = 1 \end{cases}$$

The first equation yields $a = b$ or $a = -b$. If $a = b$, then the second equation gives

$$2b^2 = 1 \implies b = \pm\frac{1}{\sqrt{2}} \implies a = \pm\frac{1}{\sqrt{2}}.$$

If $a = -b$, this results in $-2b^2 = 1$ which has no solution when $b \in \mathbb{R}$. Hence, the two distinct square roots of i are

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \quad \text{and} \quad -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i. \qquad \blacksquare$$

Exercise 1A9

Find $x \in \mathbb{R}^4$ such that

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8).$$

Solution. Let $x = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$. The equation becomes

$$(4, -3, 1, 7) + 2(x_1, x_2, x_3, x_4) = (5, 9, -6, 8),$$

which simplifies to the system of equations

$$\begin{cases} 5 = 4 + 2x_1 \\ 9 = -3 + 2x_2 \\ -6 = 1 + 2x_3 \\ 8 = 7 + 2x_4 \end{cases} \implies \begin{cases} 2x_1 = 1 \\ 2x_2 = 12 \\ 2x_3 = -7 \\ 2x_4 = 1 \end{cases}$$

Therefore,

$$x = \left(\frac{1}{2}, 6, -\frac{7}{2}, \frac{1}{2}\right). \qquad \blacksquare$$

Exercise 1A10

Explain why there does not exist $\lambda \in \mathbb{C}$ such that

$$\lambda(2-3i, 5+4i, -6+7i) = (12-5i, 7+22i, -32-9i).$$

Solution. Suppose there exists $\lambda \in \mathbb{C}$ such that

$$\lambda(2-3i, 5+4i, -6+7i) = (12-5i, 7+22i, -32-9i).$$

This gives the system of equations

$$\begin{cases} \lambda(2-3i) = 12-5i \\ \lambda(5+4i) = 7+22i \\ \lambda(-6+7i) = -32-9i \end{cases}$$

Solving the first equation for λ, we get

$$\lambda = \frac{12-5i}{2-3i} = \frac{(12-5i)(2+3i)}{(2-3i)(2+3i)} = \frac{24+36i-10i-15i^2}{4+9} = \frac{24+26i+15}{13} = 3+2i.$$

Solving the second equation for λ, we get

$$\lambda = \frac{7+22i}{5+4i} = \frac{(7+22i)(5-4i)}{(5+4i)(5-4i)} = \frac{35-28i+110i-88i^2}{25+16} = \frac{35+82i+88}{41} = 3+2i.$$

Solving the third equation for λ, we get

$$\lambda = \frac{-32-9i}{-6+7i} = \frac{(-32-9i)(-6-7i)}{(-6+7i)(-6-7i)} = \frac{192+224i+54i+63i^2}{36+49} = \frac{192+278i-63}{85} = \frac{129+278i}{85}.$$

Since the value of λ obtained from the third equation is different from the value obtained from the first two equations, no such $\lambda \in \mathbb{C}$ exists. ■

Exercise 1A11

Show that $(x+y)+z = x+(y+z)$ for all $x, y, z \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n)$, and $z = (z_1, z_2, \ldots, z_n)$. Then

$$\begin{aligned}
(x+y)+z &= ((x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n)) + (z_1, z_2, \ldots, z_n) \\
&= ((x_1+y_1), (x_2+y_2), \ldots, (x_n+y_n)) + (z_1, z_2, \ldots, z_n) \\
&= (x_1+y_1+z_1, x_2+y_2+z_2, \ldots, x_n+y_n+z_n) \\
&= (x_1, x_2, \ldots, x_n) + ((y_1+z_1), (y_2+z_2), \ldots, (y_n+z_n)) \\
&= x+(y+z).
\end{aligned}$$

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Exercise 1A12

Show that $(ab)x = a(bx)$ for all $a, b \in \mathbb{F}$ and $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{F}^n$ and $a, b \in \mathbb{F}$. Then

$$\begin{aligned}
(ab)x &= (ab)(x_1, x_2, \ldots, x_n) \\
&= ((ab)x_1, (ab)x_2, \ldots, (ab)x_n) \\
&= (a(bx_1), a(bx_2), \ldots, a(bx_n)) \\
&= a((bx_1, bx_2, \ldots, bx_n)) \\
&= a(bx).
\end{aligned}$$

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Exercise 1A13

Show that $1x = x$ for all $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \dots, x_n) \in \mathbb{F}^n$. Then

$$\begin{aligned} 1x &= 1(x_1, x_2, \dots, x_n) \\ &= (1x_1, 1x_2, \dots, 1x_n) \\ &= (x_1, x_2, \dots, x_n) \\ &= x. \end{aligned}$$

■

Exercise 1A14

Show that $\lambda(x + y) = \lambda x + \lambda y$ for all $\lambda \in \mathbb{F}$ and $x, y \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \dots, x_n)$ and $y = (y_1, y_2, \dots, y_n)$ be elements of $\mathbb{F}^n$ and $\lambda \in \mathbb{F}$. Then

$$\begin{aligned} \lambda(x + y) &= \lambda((x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n)) \\ &= \lambda((x_1 + y_1), (x_2 + y_2), \dots, (x_n + y_n)) \\ &= (\lambda(x_1 + y_1), \lambda(x_2 + y_2), \dots, \lambda(x_n + y_n)) \\ &= (\lambda x_1 + \lambda y_1, \lambda x_2 + \lambda y_2, \dots, \lambda x_n + \lambda y_n) \\ &= (\lambda x_1, \lambda x_2, \dots, \lambda x_n) + (\lambda y_1, \lambda y_2, \dots, \lambda y_n) \\ &= \lambda(x_1, x_2, \dots, x_n) + \lambda(y_1, y_2, \dots, y_n) \\ &= \lambda x + \lambda y. \end{aligned}$$

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Exercise 1A15

Show that $(a + b)x = ax + bx$ for all $a, b \in \mathbb{F}$ and $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \dots, x_n) \in \mathbb{F}^n$ and $a, b \in \mathbb{F}$. Then

$$\begin{aligned} (a + b)x &= (a + b)(x_1, x_2, \dots, x_n) \\ &= ((a + b)x_1, (a + b)x_2, \dots, (a + b)x_n) \\ &= (ax_1 + bx_1, ax_2 + bx_2, \dots, ax_n + bx_n) \\ &= (ax_1, ax_2, \dots, ax_n) + (bx_1, bx_2, \dots, bx_n) \\ &= ax + bx. \end{aligned}$$

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1B Definition of Vector Space

Exercise 1B1

Prove that $-(-v) = v$ for every $v \in V$.

Solution. Let $v \in V$. By the definition of additive inverse, we have

$$v + (-v) = 0 = -(-v) + (-v).$$

Adding $-(-v)$ to both sides, we get

$$v + (-v) + (-(-v)) = -(-v) + (-v) + (-(-v)).$$

Simplifying both sides, we obtain

$$v = -(-v).$$

■

Exercise 1B2

Suppose $a \in \mathbb{F}$, $v \in V$, and $av = 0$. Prove that $a = 0$ or $v = 0$.

Solution. If $a = 0$, then we are done. Suppose $a \neq 0$. Then

$$v = 1v = \frac{1}{a}(av) = \frac{1}{a} \cdot 0 = 0.$$

■

Exercise 1B3

Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

Solution. Solving for x explicitly, we obtain

$$3x = w - v \implies x = \frac{1}{3}(w - v).$$

This shows that there exists $x \in V$ such that $v + 3x = w$. To see that x is unique, suppose there exists another $y \in V$ such that $v + 3y = w$. Then dividing by 3, we get

$$3x = w - v = 3y \implies x = y.$$

■

Exercise 1B4

The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in the definition of a vector space. Which one?

Solution. The empty set fails to satisfy the requirement that there exists a zero vector $0 \in V$ such that $v + 0 = v$ for all $v \in V$ since it contains no elements. ■

Exercise 1B5

Show that in the definition of a vector space, the additive inverse condition can be replaced with the condition that

$$0v = 0 \text{ for all } v \in V.$$

Here, the 0 on the left side is the number 0, and the 0 on the right side is the additive identity in V.

Solution. Suppose that $0v = 0$ for all $v \in V$. Let $v \in V$. Then

$$v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0.$$

Hence, $(-1)v$ is the additive inverse of v.

Conversely, suppose that for every $v \in V$, there exists an additive inverse $-v \in V$ such that $v + (-v) = 0$. Then $0v = (0 + 0)v = 0v + 0v$. Adding $-0v$ to both sides, we get $0 = 0v$. Hence, $0v = 0$ for all $v \in V$. ∎

Exercise 1B6

Let ∞ and $-\infty$ denote two distinct objects, neither of which is in $\mathbb{R}$. Define an addition and scalar multiplication on $\mathbb{R} \cup \{\infty, -\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbb{R}$, define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \qquad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty = \infty + t = \infty + \infty &= \infty, \\ t + (-\infty) = (-\infty) + t = (-\infty) + (-\infty) &= -\infty, \\ \infty + (-\infty) = (-\infty) + \infty &= 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is $\mathbb{R} \cup \{\infty, -\infty\}$ a vector space over $\mathbb{R}$? Explain.

Solution. No, $\mathbb{R} \cup \{\infty, -\infty\}$ is not a vector space over $\mathbb{R}$ as it does not satisfy the associative property of addition. Let $t \in \mathbb{R}$. Then

$$(t + \infty) + (-\infty) = \infty + (-\infty) = 0 \neq t = t + 0 = t + (\infty + (-\infty)).$$ ∎

Exercise 1B7

Suppose S is a nonempty set. Let V^S denote the set of functions from S to V. Define a natural addition and scalar multiplication on V^S, and show that V^S is a vector space with these definitions.

Solution. Let $f, g \in V^S$ and $t \in \mathbb{F}$. Define addition and scalar multiplication as follows:

$$(f + g)(s) = f(s) + g(s), \quad (tf)(s) = t(f(s))$$

for all $s \in S$. We will show that V^S is a vector space with these operations.

1. **Commutativity**: Let $f, g \in V^S$. Then for all $s \in S$,

$$(f + g)(s) = f(s) + g(s) = g(s) + f(s) = (g + f)(s).$$

2. **Associativity**: Let $f, g, h \in V^S$ and $t, u \in \mathbb{F}$. Then for all $s \in S$,

$$\begin{aligned} ((f + g) + h)(s) &= (f + g)(s) + h(s) \\ &= (f(s) + g(s)) + h(s) \\ &= f(s) + (g(s) + h(s)) \\ &= f(s) + (g + h)(s) \\ &= (f + (g + h))(s), \end{aligned}$$

and

$$\begin{aligned}((tu)f)(s) &= (tu)(f(s)) \\ &= t(u(f(s))) \\ &= t(uf)(s) \\ &= (t(uf))(s).\end{aligned}$$

3. **Additive identity**: Let $0 \in V$ be the additive identity in V. Define the zero function $0_S \in V^S$ by $0_S(s) = 0$ for all $s \in S$. Then for all $f \in V^S$ and $s \in S$,

$$(f + 0_S)(s) = f(s) + 0_S(s) = f(s) + 0 = f(s) = 0 + f(s) = (0_S + f)(s).$$

Hence, 0_S is the additive identity in V^S.

4. **Additive inverse**: Let $f \in V^S$. Define the function $-f \in V^S$ by $(-f)(s) = -f(s)$ for all $s \in S$. Then for all $s \in S$,

$$(f + (-f))(s) = f(s) + (-f)(s) = f(s) + (-f(s)) = 0 = 0_S(s).$$

Hence, $-f$ is the additive inverse of f in V^S.

5. **Multiplicative identity**: For all $f \in V^S$ and $s \in S$,

$$(1f)(s) = 1(f(s)) = f(s).$$

6. **Distributive properties**: Let $f, g \in V^S$ and $t, u \in \mathbb{F}$. Then for all $s \in S$,

$$\begin{aligned}((t+u)f)(s) &= (t+u)(f(s)) \\ &= t(f(s)) + u(f(s)) \\ &= (tf)(s) + (uf)(s) \\ &= (tf + uf)(s),\end{aligned}$$

and

$$\begin{aligned}(t(f+g))(s) &= t((f+g)(s)) \\ &= t(f(s) + g(s)) \\ &= t(f(s)) + t(g(s)) \\ &= (tf)(s) + (tg)(s) \\ &= (tf + tg)(s).\end{aligned}$$

■

Exercise 1B8

Suppose V is a real vector space.

- The **complexification** of V, denoted by $V_{\mathbb{C}}$, equals $V \times V$. An element of $V_{\mathbb{C}}$ is an ordered pair (u, v), where $u, v \in V$, but we write this as $u + iv$.
- Addition on $V_{\mathbb{C}}$ is defined by
$$(u_1 + iv_1) + (u_2 + iv_2) = (u_1 + u_2) + i(v_1 + v_2)$$
for all $u_1, u_2, v_1, v_2 \in V$.
- Complex scalar multiplication on $V_{\mathbb{C}}$ is defined by
$$(a + ib)(u + iv) = (au - bv) + i(av + bu)$$
for all $a, b \in \mathbb{R}$ and $u, v \in V$.

Prove that with the definitions of addition and scalar multiplication as above, $V_{\mathbb{C}}$ is a complex vector space.

Solution. We prove that $V_{\mathbb{C}}$ is a complex vector space by verifying the vector space axioms.

1. **Commutativity**: Let $u_1 + v_1 i, u_2 + v_2 i \in V_{\mathbb{C}}$. Then
$$(u_1 + v_1 i) + (u_2 + v_2 i) = (u_1 + u_2) + i(v_1 + v_2) = (u_2 + u_1) + i(v_2 + v_1) = (u_2 + v_2 i) + (u_1 + v_1 i).$$

2. **Associativity**: Let $u_1 + v_1 i, u_2 + v_2 i, u_3 + v_3 i \in V_{\mathbb{C}}$. Then
$$\begin{aligned}((u_1 + v_1 i) + (u_2 + v_2 i)) + (u_3 + v_3 i) &= ((u_1 + u_2) + i(v_1 + v_2)) + (u_3 + v_3 i) \\ &= ((u_1 + u_2) + u_3) + i((v_1 + v_2) + v_3) \\ &= (u_1 + (u_2 + u_3)) + i(v_1 + (v_2 + v_3)) \\ &= (u_1 + v_1 i) + ((u_2 + v_2 i) + (u_3 + v_3 i)).\end{aligned}$$
Similarly, let $a + bi, c + di \in \mathbb{C}$ and $u + vi \in V_{\mathbb{C}}$. Then
$$\begin{aligned}((a + bi)(c + di))(u + vi) &= ((ac - bd) + i(ad + bc))(u + vi) \\ &= ((ac - bd)u - (ad + bc)v) + i((ac - bd)v + (ad + bc)u) \\ &= (a(cu - dv) - b(du + cv)) + i(a(du + cv) + b(cu - dv)) \\ &= (a + bi)((cu - dv) + i(du + cv)) \\ &= (a + bi)((c + di)(u + vi)).\end{aligned}$$

3. **Additive identity**: Let $0 \in V$ be the additive identity in V. Then for all $u + vi \in V_{\mathbb{C}}$,
$$(u + vi) + (0 + 0i) = (u + 0) + i(v + 0) = u + vi = (0 + 0i) + (u + vi).$$
Hence, $0 + 0i$ is the additive identity in $V_{\mathbb{C}}$.

4. **Additive inverse**: Let $u + vi \in V_{\mathbb{C}}$. Then
$$(u + vi) + (-u + (-v)i) = (u - u) + i(v - v) = 0 + 0i.$$
Hence, $-u + (-v)i$ is the additive inverse of $u + vi$ in $V_{\mathbb{C}}$.

5. **Multiplicative identity**: For all $u + vi \in V_{\mathbb{C}}$,

$$(1 + 0i)(u + vi) = (1u - 0v) + i(1v + 0u) = u + vi.$$

   Hence, $1 + 0i$ is the multiplicative identity in $V_{\mathbb{C}}$.

6. **Distributive properties**: Let $a + bi, c + di \in \mathbb{C}$ and $u_1 + v_1 i, u_2 + v_2 i \in V_{\mathbb{C}}$. Then

$$\begin{aligned}(a+bi)((u_1+v_1 i)+(u_2+v_2 i)) &= (a+bi)((u_1+u_2)+i(v_1+v_2)) \\ &= (a(u_1+u_2)-b(v_1+v_2))+i(a(v_1+v_2)+b(u_1+u_2)) \\ &= (au_1-bv_1)+i(av_1+bu_1)+(au_2-bv_2)+i(av_2+bu_2) \\ &= (a+bi)(u_1+v_1 i)+(a+bi)(u_2+v_2 i),\end{aligned}$$

   and

$$\begin{aligned}((a+bi)+(c+di))(u_1+v_1 i) &= ((a+c)+(b+d)i)(u_1+v_1 i) \\ &= ((a+c)u_1-(b+d)v_1)+i((a+c)v_1+(b+d)u_1) \\ &= (au_1-bv_1)+i(av_1+bu_1)+(cu_1-dv_1)+i(cv_1+du_1) \\ &= (a+bi)(u_1+v_1 i)+(c+di)(u_1+v_1 i).\end{aligned}$$

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1C Subspaces

Exercise 1C1

For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$.

(a) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 + 2x_2 + 3x_3 = 0\}$

(b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 + 2x_2 + 3x_3 = 4\}$

(c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 x_2 x_3 = 0\}$

(d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 = 5x_3\}$

Solution. Let U be the subset of $\mathbb{F}^3$ in question.

(a) Since the zero vector $(0, 0, 0)$ satisfies $0 + 2(0) + 3(0) = 0$, U is nonempty. Let $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$ be elements of U. Then

$$(u_1 + v_1) + 2(u_2 + v_2) + 3(u_3 + v_3) = (u_1 + 2u_2 + 3u_3) + (v_1 + 2v_2 + 3v_3) = 0 + 0 = 0.$$

Hence, $u + v$ is in U. Let $a \in \mathbb{F}$. Then

$$au_1 + 2(au_2) + 3(au_3) = a(u_1 + 2u_2 + 3u_3) = a \cdot 0 = 0.$$

Hence, au is in U. Therefore, U is a subspace of $\mathbb{F}^3$.

(b) The zero vector $(0, 0, 0)$ does not satisfy $0 + 2(0) + 3(0) = 4$, so U does not contain the zero vector. Hence, it is not a subspace of $\mathbb{F}^3$.

(c) Suppose $u = (1, 0, 0)$ and $v = (0, 1, 1)$. Then u and v are in U since $1 \cdot 0 \cdot 0 = 0$ and $0 \cdot 1 \cdot 1 = 0$. However, $u + v = (1, 1, 1)$ is not in U since $1 \cdot 1 \cdot 1 = 1 \neq 0$. Hence, U is not closed under addition and is not a subspace of $\mathbb{F}^3$.

(d) Since the zero vector $(0, 0, 0)$ satisfies $0 = 5(0)$, U is nonempty. Let $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$ be elements of U. Then

$$u_1 + v_1 = 5u_3 + 5v_3 = 5(u_3 + v_3).$$

Hence, $u + v$ is in U. Let $a \in \mathbb{F}$. Then

$$au_1 = a(5u_3) = 5(au_3).$$

Hence, au is in U. Therefore, U is a subspace of $\mathbb{F}^3$. ∎

Exercise 1C2

Verify all assertions about subspaces in Example 1.35.

(a) If $b \in \mathbb{F}$, then

$$\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_3 = 5x_4 + b\}$$

is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

(b) The set of continuous, real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.

(c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$.

(d) The set of differentiable real-valued functions f on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

(e) The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbb{C}^{\infty}$.

Solution. Let U be the subset of $\mathbb{F}^4$ in question.

(a) Suppose U is a subspace of $\mathbb{F}^4$. Then it must contain the zero vector $(0, 0, 0, 0)$. Then $0 = 5(0) + b$, which implies $b = 0$.

Conversely, suppose $b = 0$. Then $U = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_3 = 5x_4\}$. The proof is similar to that of Exercise 1C1 (d), hence U is a subspace of $\mathbb{F}^4$.

(b) Note that the 0 function on $[0, 1]$ is continuous, so $0 \in U$. Let $f, g \in U$ be continuous functions. Then $f + g$ is continuous, so $f + g \in U$. Let $a \in \mathbb{R}$. Then af is continuous, so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{[0,1]}$.

(c) Since the derivative of the 0 function is the 0 function, $0 \in U$. Let $f, g \in U$ be differentiable functions. Then $f + g$ is differentiable, so $f + g \in U$. Let $a \in \mathbb{R}$. Then af is differentiable, so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{\mathbb{R}}$.

(d) Suppose U is a subspace of $\mathbb{R}^{(0,3)}$. Then it must contain the 0 function on $(0, 3)$. Then $0' = 0 = b$, which implies $b = 0$.

Conversely, suppose $b = 0$. Then $U = \{f \in \mathbb{R}^{(0,3)} \mid f'(2) = 0\}$. Let $f, g \in U$ be differentiable functions such that $f'(2) = g'(2) = 0$. Then

$$(f + g)'(2) = f'(2) + g'(2) = 0 + 0 = 0,$$

so $f + g \in U$. Let $a \in \mathbb{R}$. Then

$$(af)'(2) = af'(2) = a \cdot 0 = 0,$$

so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{(0,3)}$.

(e) Let U be the set of all sequences of complex numbers with limit 0. Then the zero sequence is in U. Let $(z_n), (w_n) \in U$ be sequences with limit 0. Then

$$\lim_{n\to\infty} (z_n + w_n) = \lim_{n\to\infty} z_n + \lim_{n\to\infty} w_n = 0 + 0 = 0,$$

so $(z_n) + (w_n) \in U$. Let $a \in \mathbb{C}$. Then

$$\lim_{n\to\infty} (az_n) = a \lim_{n\to\infty} z_n = a \cdot 0 = 0,$$

so $a(z_n) \in U$. Therefore, U is a subspace of $\mathbb{C}^\infty$. ■

Exercise 1C3

Show that the set of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbb{R}^{(-4,4)}$.

Solution. Let U be the set in question. Then the zero function is in U since $0'(-1) = 0 = 3 \cdot 0 = 3 \cdot 0(2)$. Let $f, g \in U$ be differentiable functions such that $f'(-1) = 3f(2)$ and $g'(-1) = 3g(2)$. Then

$$(f + g)'(-1) = f'(-1) + g'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3(f + g)(2),$$

so $f + g \in U$. Let $a \in \mathbb{R}$. Then

$$(af)'(-1) = af'(-1) = a \cdot 3f(2) = 3(af)(2),$$

so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{(-4,4)}$. ■

Exercise 1C4

Suppose $b \in \mathbb{R}$. Show that the set of continuous real-valued functions f on the interval $[0, 1]$ such that

$$\int_0^1 f = b$$

is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Solution. Let U be the set in question. Suppose U is a subspace of $\mathbb{R}^{[0,1]}$. Then it must contain the zero function on $[0, 1]$. Then

$$\int_0^1 0 = 0 = b,$$

which implies $b = 0$.

Conversely, suppose $b = 0$. Then

$$U = \left\{ f \in \mathbb{R}^{[0,1]} \,\middle|\, \int_0^1 f = 0 \right\}.$$

The zero function is in U since $\int_0^1 0 = 0$. Let $f, g \in U$ be continuous functions such that $\int_0^1 f = \int_0^1 g = 0$. Then

$$\int_0^1 (f + g) = \int_0^1 f + \int_0^1 g = 0 + 0 = 0,$$

so $f + g \in U$. Let $a \in \mathbb{R}$. Then

$$\int_0^1 (af) = a \int_0^1 f = a \cdot 0 = 0,$$

so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{[0,1]}$. ■

Exercise 1C5

Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Solution. No, $\mathbb{R}^2$ is not a subspace of $\mathbb{C}^2$. Let $u = (1, 0) \in \mathbb{R}^2$ and $a = i \in \mathbb{C}$. Then $au = (i, 0) \notin \mathbb{R}^2$, so $\mathbb{R}^2$ is not closed under scalar multiplication and is not a subspace of $\mathbb{C}^2$. ■

Exercise 1C6

(a) Is $\{(a, b, c) \in \mathbb{R}^3 \mid a^3 = b^3\}$ a subspace of $\mathbb{R}^3$?

(b) Is $\{(a, b, c) \in \mathbb{C}^3 \mid a^3 = b^3\}$ a subspace of $\mathbb{C}^3$?

Remark. Here, I present an elementary argument that does not require the use of De Moivre's Theorem; rather, I arrive to the conclusion directly. However, readers who are familiar with it may find it easier to note that the complex number 1 has three cubic roots of unity. Other readers may also use Vieta's formulas to arrive at the same polynomial in ω to obtain $1 + \omega = -\omega^2$.

Solution. Let U be the subset in question.

(a) Since the zero vector $(0, 0, 0)$ satisfies $0^3 = 0^3$, U is nonempty. Let $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ be elements of U. Then $x_1^3 = x_2^3$ and $y_1^3 = y_2^3$ both imply that $x_1 = x_2$ and $y_1 = y_2$. Hence,

$$(x_1 + y_1)^3 = (x_2 + y_2)^3,$$

so $x + y \in U$. Let $a \in \mathbb{R}$. Then

$$(ax_1)^3 = a^3x_1^3 = a^3x_2^3 = (ax_2)^3,$$

so $ax \in U$. Therefore, U is a subspace of $\mathbb{R}^3$.

(b) No, U is not a subspace of $\mathbb{C}^3$. To see this, consider the cubic polynomial $x^3 - 1 = 0$. This polynomial has three distinct roots in $\mathbb{C}$ as follows:

$$x^3 - 1 = (x - 1)(x^2 + x + 1) = 0$$

Applying the quadratic formula to $x^2 + x + 1 = 0$, we get

$$x = \frac{-1 \pm \sqrt{(-1)^2 - 4(1)(1)}}{2(1)} = \frac{-1 \pm i\sqrt{3}}{2}.$$

Let

$$\omega = \frac{-1 + i\sqrt{3}}{2}.$$

Then $\omega^3 = 1$, and

$$\omega^2 = \frac{-1 - i\sqrt{3}}{2}.$$

Moreover, it is not hard to see that $\omega + \omega^2 = -1$. Let $u = (1, 1, 0)$ and $v = (\omega, 1, 0)$. Then $u, v \in U$ since $1^3 = 1^3$ and $\omega^3 = 1^3$. Then $u + v = (1 + \omega, 2, 0)$. However,

$$(1 + \omega)^3 = 1 + 3\omega + 3\omega^2 + \omega^3 = 1 + 3(-1) + 1 = -1 \neq 2^3 = 8.$$

Hence, $u + v \notin U$, so U is not closed under addition and is not a subspace of $\mathbb{C}^3$. ∎

Exercise 1C7

Prove or give a counterexample: If U is a nonempty subset of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbb{R}^2$.

Solution. Consider the subset $\mathbb{Q}^2$ of $\mathbb{R}^2$. Then $\mathbb{Q}^2$ is nonempty because $(0, 0) \in \mathbb{Q}^2$, closed under addition since the sum of any two rational numbers is rational, and closed under taking additive inverses since the additive inverse of a rational number is rational. However, for any irrational number $r \in \mathbb{R}$, the vector $(r, 0) \in \mathbb{R}^2$ is not in $\mathbb{Q}^2$. Hence, $\mathbb{Q}^2$ is not closed under scalar multiplication and is not a subspace of $\mathbb{R}^2$. Therefore, the statement is false. ∎

Exercise 1C8

Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbb{R}^2$.

Solution. Consider the subset

$$U = \{(x, 0) \mid x \in \mathbb{R}\} \cup \{(0, y) \mid y \in \mathbb{R}\}.$$

Then U is nonempty since $(1, 0) \in U$, and it is closed under scalar multiplication since any scalar multiple of $(1, 0)$ or $(0, 1)$ is still in U. However, U is not closed under addition since $(1, 0) + (0, 1) = (1, 1) \notin U$. Hence, U is not a subspace of $\mathbb{R}^2$. ∎

Exercise 1C9

A function $f : \mathbb{R} \to \mathbb{R}$ is called *periodic* if there exists a positive number p such that $f(x + p) = f(x)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$? Explain.

Solution. It is not a subspace of $\mathbb{R}^{\mathbb{R}}$. Consider the periodic functions $f(x) = \sin(2x)$ and $g(x) = \sin(\pi x)$. The function f has period π, and the function g has period 2. For $f(x + p) + g(x + p) = f(x) + g(x)$ to hold for all $x \in \mathbb{R}$, we would need a common period p that is a multiple of both π and 2. However, no such positive number exists since π is irrational. Therefore, the sum $f + g$ is not periodic, and the set of periodic functions is not closed under addition. Hence, it is not a subspace of $\mathbb{R}^{\mathbb{R}}$. ∎

Exercise 1C10

Suppose V_1 and V_2 are subspaces of V. Prove that the intersection $V_1 \cap V_2$ is a subspace of V.

Solution. Let $U = V_1 \cap V_2$. Since both V_1 and V_2 are subspaces of V, they both contain the zero vector of V. Hence, $0 \in U$, so U is nonempty. Let $u, v \in U$. Then $u, v \in V_1$ and $u, v \in V_2$. Since both V_1 and V_2 are subspaces, they are closed under addition. Therefore, $u + v \in V_1$ and $u + v \in V_2$, which implies that $u + v \in U$. Let $a \in \mathbb{F}$. Then since both V_1 and V_2 are closed under scalar multiplication, we have that $au \in V_1$ and $au \in V_2$, which implies that $au \in U$. Therefore, U is a subspace of V. ∎

Exercise 1C11

Prove that the intersection of every collection of subspaces of V is a subspace of V.

Solution. Let $\mathcal{C}$ be a collection of subspaces of V, and let

$$U = \bigcap_{W \in \mathcal{C}} W.$$

Since each subspace in $\mathcal{C}$ contains the zero vector, $0 \in U$, so U is nonempty. Let $u, v \in U$. Then $u, v \in W$ for all $W \in \mathcal{C}$. Since each W is a subspace, they are closed under addition, so $u + v \in W$ for all $W \in \mathcal{C}$, which implies $u + v \in U$. Let $a \in \mathbb{F}$. Then $au \in W$ for all $W \in \mathcal{C}$, which implies $au \in U$. Therefore, U is a subspace of V. ∎

Exercise 1C12

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Solution. Let U, W be subspaces of V. Suppose $U \subseteq W$. Then $U \cup W = W$, which is a subspace of V. Similarly, if $W \subseteq U$, then $U \cup W = U$, which is a subspace of V.

Conversely, suppose $U \cup W$ is a subspace of V. If neither $U \subseteq W$ nor $W \subseteq U$, then there exist $u \in U \setminus W$ and $w \in W \setminus U$. Since $U \cup W$ is a subspace, it must be closed under addition. Then $u + w \in U \cup W$. However, $u + w \notin U$ because if it were, then $w = (u + w) - u \in U$, which is a contradiction. Similarly, $u + w \notin W$ because if it were, then $u = (u + w) - w \in W$, which is also a contradiction. Hence, $u + w \notin U \cup W$, which contradicts the assumption that $U \cup W$ is a subspace. Therefore, one of the subspaces must be contained in the other. ∎

Exercise 1C13

Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

Solution. Let U_1, U_2, U_3 be subspaces of V. Without loss of generality, suppose $U_2 \subseteq U_1$ and $U_3 \subseteq U_1$. Then $U_1 \cup U_2 \cup U_3 = U_1$, which is a subspace of V.

Conversely, suppose $U_1 \cup U_2 \cup U_3$ is a subspace of V. Note that $U_1 \cup U_2 \cup U_3 = (U_1 \cup U_2) \cup U_3$. Using Exercise 1C12, one of the subspaces $U_1 \cup U_2$ or U_3 must be contained in the other. If $U_1 \cup U_2 \subseteq U_3$, then $U_1 \cup U_2 \cup U_3 = U_3$, which is a subspace of V. If $U_3 \subseteq U_1 \cup U_2$, then U_3 must be contained in either U_1 or U_2.

Without loss of generality, suppose $U_3 \subseteq U_1$. Then $U_1 \cup U_2 \cup U_3 = U_1 \cup U_2$. Using Exercise 1C12 again, one of the subspaces U_1 or U_2 must be contained in the other. If $U_2 \subseteq U_1$, then $U_1 \cup U_2 \cup U_3 = U_1$, which is a subspace of V. If $U_1 \subseteq U_2$, then $U_1 \cup U_2 \cup U_3 = U_2$, which is a subspace of V. Therefore, one of the subspaces must contain the other two. ∎

Exercise 1C14

Suppose
$$U = \{(x, -x, 2x) \in \mathbb{F}^3 \mid x \in \mathbb{F}\} \quad \text{and} \quad W = \{(x, x, 2x) \in \mathbb{F}^3 \mid x \in \mathbb{F}\}.$$
Describe $U + W$ using symbols, and also give a description of $U + W$ that uses no symbols.

Solution. Let $u = (x, -x, 2x) \in U$ and $w = (y, y, 2y) \in W$. Then
$$u + w = (x + y, -x + y, 2x + 2y) = (x + y, y - x, 2(x + y)).$$
Observe that $x + y$ can take any value in $\mathbb{F}$, and $y - x$ can also take any value in $\mathbb{F}$. Hence, we can write
$$U + W = \{(a, b, 2a) \in \mathbb{F}^3 \mid a, b \in \mathbb{F}\}.$$
In words, $U + W$ is the set of all vectors in $\mathbb{F}^3$ whose third component is twice the first component, and whose second component can be any value in $\mathbb{F}$. ■

Exercise 1C15

Suppose U is a subspace of V. What is $U + U$?

Solution. Let $u_1, u_2 \in U$. Then $u_1 + u_2 \in U$ since U is closed under addition. Hence, $U + U \subseteq U$. Conversely, let $u \in U$. Then $u = u + 0$, where 0 is the additive identity in U. Since $0 \in U$, we have that $u \in U + U$. Hence, $U \subseteq U + U$. Therefore, $U + U = U$. ■

Exercise 1C16

Is the operation of addition on subspaces of V commutative? In other words, if U and W are subspaces of V, is $U + W = W + U$?

Solution. Yes, the operation of addition on subspaces of V is commutative. Let $u \in U$ and $w \in W$. Then $u + w \in U + W$. Because addition in V is commutative, we have that $u + w = w + u$, where $w \in W$ and $u \in U$. Hence, $w + u \in W + U$. Therefore, every element of $U + W$ is also an element of $W + U$, which implies that $U + W \subseteq W + U$. By a similar argument, we can show that $W + U \subseteq U + W$. Hence, $U + W = W + U$. ■

Exercise 1C17

Is the operation of addition on the subspaces of V associative? In other words, if V_1, V_2, V_3 are subspaces of V, is
$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

Solution. Yes, the operation of addition on subspaces of V is associative. Let $u_1 \in V_1$, $u_2 \in V_2$, and $u_3 \in V_3$. Then $(u_1 + u_2) + u_3 \in (V_1 + V_2) + V_3$. Because addition in V is associative, we have that $(u_1 + u_2) + u_3 = u_1 + (u_2 + u_3)$, where $u_1 \in V_1$, $u_2 \in V_2$, and $u_3 \in V_3$. Hence, $u_1 + (u_2 + u_3) \in V_1 + (V_2 + V_3)$. Therefore, every element of $(V_1+V_2)+V_3$ is also an element of $V_1+(V_2+V_3)$, which implies that $(V_1+V_2)+V_3 \subseteq V_1+(V_2+V_3)$. By a similar argument, we can show that $V_1+(V_2+V_3) \subseteq (V_1+V_2)+V_3$. Hence, $(V_1+V_2)+V_3 = V_1+(V_2+V_3)$. ■

Exercise 1C18

Does the operation of addition on the subspaces of V have an additive identity? Which subspaces have additive inverses?

Solution. Yes, the operation of addition on the subspaces of V has an additive identity. The additive identity is the zero subspace $\{0\}$, which contains only the zero vector of V. For any subspace U of V, we have that $U + \{0\} = U$.

Every subspace U of V has an additive inverse, which is itself. For any subspace U, we have that $U+U = U$, as shown in Exercise 1C15. Therefore, every subspace is its own additive inverse under the operation of addition on subspaces. ∎

Exercise 1C19

Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V_1 + U = V_2 + U,$$

then $V_1 = V_2$.

Solution. The statement is false. Consider the vector space $\mathbb{R}^2$ and let

$$V_1 = \{(x, 0) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}, \quad V_2 = \{(0, y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\}, \quad U = \{(x, x) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}.$$

Then $V_1 + U = \mathbb{R}^2$ and $V_2 + U = \mathbb{R}^2$, but $V_1 \neq V_2$. Therefore, the statement is false. ∎

Exercise 1C20

Suppose

$$U = \{(x, x, y, y) \in \mathbb{F}^4 \mid x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^4$ such that $\mathbb{F}^4 = U \oplus W$.

Solution. Let

$$W = \{(0, 0, z, w) \in \mathbb{F}^4 \mid z, w \in \mathbb{F}\}.$$

If $(x, x, y, y) \in U$ and $(0, 0, z, w) \in W$, then $(x, x, y, y) = (0, 0, z, w)$ if and only if $x = 0$, $y = 0$, $z = 0$, and $w = 0$. Hence, $U \cap W = \{0\}$. Moreover, for any $(a, b, c, d) \in \mathbb{F}^4$, we can write

$$(a, b, c, d) = (x, x, y, y) + (0, 0, z, w),$$

where

$$x = a, \quad y = c, \quad z = c - y = 0, \quad w = d - y = d - c.$$

Hence, every vector in $\mathbb{F}^4$ can be written as a sum of a vector in U and a vector in W. Therefore, $\mathbb{F}^4 = U \oplus W$. ∎

Exercise 1C21

Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 \mid x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^5$ such that $\mathbb{F}^5 = U \oplus W$.

Solution. Let

$$W = \{(0, 0, z, w, v) \in \mathbb{F}^5 \mid z, w, v \in \mathbb{F}\}.$$

If $(x, y, x + y, x - y, 2x) \in U$ and $(0, 0, z, w, v) \in W$, then $(x, y, x + y, x - y, 2x) = (0, 0, z, w, v)$ if and only if $x = 0$, $y = 0$, $z = 0$, $w = 0$, and $v = 0$. Hence, $U \cap W = \{0\}$. Moreover, for any $(a, b, c, d, e) \in \mathbb{F}^5$, we can write

$$(a, b, c, d, e) = (x, y, x + y, x - y, 2x) + (0, 0, z, w, v),$$

which implies

$$\begin{cases} a = x + 0 \\ b = y + 0 \\ c = (x + y) + z \\ d = (x - y) + w \\ e = 2x + v \end{cases} \implies \begin{cases} x = a \\ y = b \\ z = c - (x + y) = c - (a + b) \\ w = d - (x - y) = d - (a - b) \\ v = e - 2x = e - 2a \end{cases}$$

Hence, every vector in $\mathbb{F}^5$ can be written as a sum of a vector in U and a vector in W. Therefore, $\mathbb{F}^5 = U \oplus W$. ∎

Exercise 1C22

Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 \mid x, y \in \mathbb{F}\}.$$

Find three subspaces W_1, W_2, W_3 of $\mathbb{F}^5$, none of which equals $\{0\}$, such that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.

Solution. Let

$$\begin{aligned} W_1 &= \{(0, 0, z, 0, 0) \in \mathbb{F}^5 \mid z \in \mathbb{F}\}, \\ W_2 &= \{(0, 0, 0, w, 0) \in \mathbb{F}^5 \mid w \in \mathbb{F}\}, \\ W_3 &= \{(0, 0, 0, 0, v) \in \mathbb{F}^5 \mid v \in \mathbb{F}\}. \end{aligned}$$

The intersection of any two of these subspaces is $\{0\}$, since they have no nonzero vectors in common. Additionally, the intersection of W_1, W_2, W_3 with U is also $\{0\}$, since any nonzero vector in U has nonzero entries in the first two coordinates, while any vector in W_1, W_2, or W_3 has zeros in the first two coordinates. The proof of showing that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$ is similar to that of Exercise 1C21, hence omitted. Therefore, $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$. ■

Exercise 1C23

Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V = V_1 \oplus U \quad \text{and} \quad V = V_2 \oplus U,$$

then $V_1 = V_2$.

Solution. See Exercise 1C19 for a counterexample. For $V_1 \oplus U$, observe that $V_1 \cap U = \{0\}$. Moreover, for any $(a, b) \in \mathbb{R}^2$, we can write

$$(a, b) = (a - b, 0) + (b, b) \in V_1 \oplus U.$$

The proof for $V_2 \oplus U$ is similar. Hence, $V = V_1 \oplus U = V_2 \oplus U$, but $V_1 \neq V_2$. Therefore, the statement is false. ■

Exercise 1C24

A function $f : \mathbb{R} \to \mathbb{R}$ is called *even* if

$$f(-x) = f(x)$$

for all $x \in \mathbb{R}$. A function $f : \mathbb{R} \to \mathbb{R}$ is called *odd* if

$$f(-x) = -f(x)$$

for all $x \in \mathbb{R}$. Let V_e denote the set of real-valued even functions on $\mathbb{R}$, and let V_o denote the set of real-valued odd functions on $\mathbb{R}$. Show that $\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$.

Solution. Let $f \in \mathbb{R}^{\mathbb{R}}$. Define

$$f_e(x) = \frac{f(x) + f(-x)}{2} \quad \text{and} \quad f_o(x) = \frac{f(x) - f(-x)}{2}.$$

Then f_e is even since

$$f_e(-x) = \frac{f(-x) + f(x)}{2} = f_e(x),$$

and f_o is odd since

$$f_o(-x) = \frac{f(-x) - f(x)}{2} = -\frac{f(x) - f(-x)}{2} = -f_o(x).$$

Moreover, we have that

$$f(x) = f_e(x) + f_o(x).$$

Hence, every function in $\mathbb{R}^{\mathbb{R}}$ can be written as a sum of an even function and an odd function.

Now, suppose $g \in V_e \cap V_o$. Then g is both even and odd, which implies that

$$g(0) = g(-0) = -g(0),$$

so $g(0) = 0$. For any $x \in \mathbb{R}$, we have that

$$g(x) = g(-(-x)) = -g(-x) = -g(x),$$

which implies that $g(x) = 0$. Hence, the only function in the intersection of V_e and V_o is the zero function. Therefore, $\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$. ∎